

Cloud Computing & DevOps – AWS Handbook

SESSION 13

CI/CD with AWS CodePipeline & CodeBuild

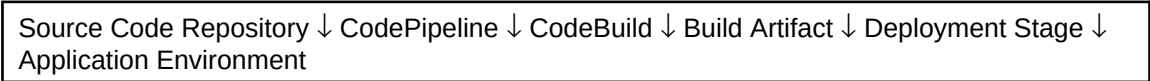
Automating Software Delivery

Duration	2 Hours
Prerequisites	Git Basics, Docker, Kubernetes Fundamentals
Learning Outcome	Understand Continuous Integration and Continuous Delivery, create build projects, and automate deployment pipelines using AWS services.

Why Are We Learning This?

Modern software development requires rapid and reliable delivery of application updates. Continuous Integration (CI) and Continuous Delivery (CD) automate the build, test, and deployment process. AWS CodeBuild and CodePipeline help organizations implement DevOps practices and reduce manual effort.

Architecture Diagram



Key Concepts

CI	Continuous Integration of code changes.
CD	Continuous Delivery and deployment automation.
CodeBuild	AWS managed build service.
CodePipeline	AWS managed CI/CD pipeline service.
Build Artifact	Output generated after build process.
Pipeline	Automated sequence of delivery stages.

Practical Activities

- Create a source code repository connection
- Create an AWS CodeBuild project
- Configure build specifications
- Run a build process
- Create an AWS CodePipeline pipeline
- Connect source, build and deployment stages
- Verify successful pipeline execution

Expected Outputs

CodeBuild project created successfully, build execution completed successfully, CodePipeline stages executed automatically, and deployment workflow completed.

What You Should See

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CodeBuild Project Configuration

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Successful Build Execution

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CodePipeline Dashboard

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Pipeline Stage Execution

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Successful Deployment Output

Journal Writing Template

Aim

To automate software build and deployment processes using AWS CodeBuild and CodePipeline.

Theory

CI/CD automates software delivery by integrating code changes, building applications, and deploying updates through automated workflows.

Procedure

Write 5–7 concise steps performed during the practical session.

Observations

Builds executed successfully, pipeline stages completed automatically, and deployments occurred without manual intervention.

Conclusion

AWS CodeBuild and CodePipeline simplified software delivery and demonstrated core DevOps automation principles.

Viva Preparation

1. What is Continuous Integration (CI)?
2. What is Continuous Delivery (CD)?
3. What is AWS CodeBuild?
4. What is AWS CodePipeline?
5. What is a Build Artifact?
6. What is a Deployment Pipeline?
7. Why is CI/CD important in DevOps?

Troubleshooting Box

Problem	Possible Cause
Build failed	Build specification or source code issue
Pipeline not triggered	Source repository integration issue
Deployment failed	Configuration error in deployment stage
Artifact unavailable	Build output generation issue

Industry Relevance

CI/CD pipelines are a core component of modern DevOps practices. Organizations use automated pipelines to accelerate software delivery, improve reliability, and reduce deployment risks.

Session Summary

Students learned CI/CD fundamentals, build automation, pipeline creation, and software delivery automation using AWS CodeBuild and CodePipeline.